

Personal Baselines: Not Every Normal is the Same

A personal baseline sounds like it should just be an average. Collect a few weeks of data, work out the middle, and call that the user's "normal". Unfortunately physiology is a bit more annoying than that. A useful baseline needs to answer 3 questions: what is typical for this person, how much do they normally vary, and has their "normal" actually changed?

The problem becomes way more obvious when these wearable metrics are put beside each other: sleep duration, sleep start time, sleep end time, sleep efficiency, heart rate variability (HRV), resting heart rate (RHR), skin temperature variation, respiratory rate, oxygen saturation and strain load. They do not all behave in the same way. HRV is noisy and skewed, sleep times wrap around midnight, oxygen saturation sits close to a hard ceiling, and strain is supposed to jump around depending on the training plan. One shared baseline calculation is mathematically easy, but likely to be physiologically wrong.

We find that personal comparison is particularly strong for RHR. Daily RHR varies substantially between people while remaining much more characteristic within the same person, although age, sex, sleep and time of year still influence it. A long-term personal view is therefore more useful than asking whether everybody should have the same RHR. [1] HRV depends even more heavily on the individual and the measurement conditions. Work using Ln rMSSD (a common way of measuring HRV) found that averaging several days gives a more stable view than trusting one daily value, with much of the improvement appearing after roughly 3 to 4 measurements. [2] A single morning can still matter, although a short trend usually carries more evidence than one weird number.

The measurement setup also belongs in the baseline record as this also matters. Sleep HRV and a random daytime measurement are not interchangeable, and a value from one device or HRV method should not quietly continue the baseline from another. Validation work shows wearable performance differs by device and metric, even when the devices are used in seemingly similar conditions. [3]

Sleep creates a separate set of problems because sleep start and end are circular values: 23:50 and 00:10 are only 20 minutes apart, while a normal arithmetic average treats them as opposite ends of the day and can place the "average" around midday (which is obviously not useful). Circular statistics, or deviations around a chosen anchor, are a better fit. Weekday and weekend patterns also need separating so a regular weekend lie-in is not mistaken for a random anomaly.

Sleep regularity then also adds another choice. The Sleep Regularity Index, Composite Phase Deviation, standard deviation of sleep timing and social jetlag each capture slightly different behaviour. Naps, awakenings, all-nighters and the length of the measurement period affect them differently. [4] A baseline probably doesn't need every measure, but it does need a clear target: consistency with the previous day, consistency with the user's longer-term schedule, or distance from their likely circadian preference.

Wearable sleep measurements also have their own bias. Recent systematic review evidence suggests consumer wearables tend to overestimate total sleep time and sleep efficiency and underestimate wake after sleep onset, with meaningful differences between devices. [5] These

wearables are imperfect, but still capture useful personal trends when used consistently. A device change should therefore trigger a new calibration period, and narrow-looking sleep-efficiency ranges shouldn't be treated as lab truth.

Population guidance shouldn't disappear just because the app is personalised. Adults are generally recommended to sleep at least 7 hours regularly, even though individual need varies. [6] If someone has slept consistently for 5.5 hours for the last 2 months, their personal average should not then teach the app that 5.5 hours is now ideal. For sleep duration we need both a personal baseline, describing what the user usually does, and an adequacy target, describing whether that pattern is likely to meet their needs. "Normal for you" and "good for you" are not always the same thing.

From this, the most sensible starting structure for each metric looks like this:

- Sleep duration: The nightly baseline should come from the main sleep session, with naps tracked separately. A 28-day period is reasonable for describing habitual sleep, but work-days and free days should remain distinguishable. Each night can then be compared with both the personal baseline and an independently estimated sleep need. Chronically short sleep should never redefine adequacy.
- Sleep start and sleep end: Since bedtimes wrap around midnight, circular calculations are a better fit than a median of minutes after midnight. Bedtime, wake time and potentially midsleep should be stored separately, then compared with the recent schedule and the corresponding day type. Travel or timezone changes temporarily reduce confidence; they do not suddenly create a new bedtime.
- Sleep efficiency: Small nightly changes are often noise, so a provider-specific rolling median and robust spread should be more reliable than reacting immediately. Efficiency and awake time also come from the same sleep/wake classification, which limits how independent they really are. After a device change, the baseline should reset or be bridged through an overlap study.
- HRV: Comparability matters more than quantity. Overnight or controlled-rest measurements can be combined when they use the same device, conditions and HRV definition. Where the source is rMSSD, its natural log is preferable because raw HRV is usually skewed. A longer personal reference can sit alongside a 3 to 7-day recent average. One low night may influence readiness, but a persistent fall beyond the person's expected variation should matter much more.
- Resting heart rate: A consistent resting or overnight measurement window should anchor this baseline. A 28 to 60-day reference with a shorter recent trend is sensible. RHR generally needs less transformation than HRV, although illness, heat, altitude and device changes can still create real shifts. Those shifts belong in the context record before they are absorbed into "normal". [1]
- Skin temperature variation: Some providers already report this metric as a change from their own temperature baseline. Creating another baseline without understanding that definition risks measuring a baseline of a baseline. Like-for-like overnight measurements are the priority, with cycle-aware reference ranges where possible. Sustained deviations carry more weight than one warm night.
- Respiratory rate: The main sleep period gives the most consistent reference, ideally using the same device and method throughout. Wearable-derived respiratory rate can agree well with reference sleep data, although the estimate remains method-dependent. [8]

The metric is relatively stable within an individual, making a small persistent rise more meaningful than a wide population threshold. One elevated night is a warning signal; several concordant nights deserve more attention.

- Oxygen saturation: A conventional symmetric “normal range” is a poor fit here. SpO₂ is bounded near 100%, the clinically interesting direction is mostly downward, and device error is not uniform. Systematic reviews have also found accuracy differences associated with skin pigmentation. [9] Quality-filtered overnight summaries should track the person’s usual level and lower tail while retaining sensible absolute safety thresholds. Altitude and device source remain part of the context.
- Strain load: Genuine rest days belong at zero. Missing wear remains unknown (there’s only so much you can do without the data). Non-wear can bias even simple dashboards because it is often related to what the person was actually doing. [12] Strain also has a weekly rhythm, so a 28-day rolling mean may flatten useful structure. Acute load, a longer exponentially weighted history and day-of-week context should be kept separately. Ratios can describe how recent load differs from prior load, but the literature does not support an acute:chronic ratio as a proven injury-risk switch. [11] [13]

Baseline estimation and change detection should be treated as separate jobs.

The stable reference moves slowly and should survive the same illness, travel period or overload block that the system is trying to detect. Change detection runs on a faster timescale, comparing the latest value and recent trend with that reference while accounting for measurement quality and context.

A useful system would recognise 3 kinds of change:

- A single-day deviation: possibly real, but still likely to contain noise.
- A short sustained deviation: several comparable readings moving in the same direction, which is much stronger evidence of a temporary physiological state.
- A structural change: a new level that persists after the temporary context has passed, such as improved fitness lowering RHR or a genuinely changed sleep schedule.

The baseline should adapt slowly. Short-term fluctuations belong in the recent trend, while sustained changes are gradually incorporated into the long-term reference. A trend that rises and then returns should barely move the reference; a genuinely stable new level should eventually become part of it.

One metric changing in isolation is usually not convincing. Temperature, HRV, heart rate, breathing rate and sleep efficiency moving together are much harder to dismiss as noise. Wearable studies have detected infections from combinations of these signals relative to individual baselines, our app obviously should not diagnose illness (obviously). [10] However this shows that 3 related changes carry more evidence than 1 outlier. Correlated signals still need careful handling so the same underlying change is not counted several times as independent evidence.

Missing data should stay missing. A day without a watch isn’t necessarily a rest day, a night without stages is not zero REM, and a missing SpO₂ value is not normal oxygen saturation. Valid-day count, recency, source consistency and coverage should travel with the baseline. Confidence describes how much evidence supports it; confidence does not make the user’s physiological state better or worse.

Taken together, this points to the following general baseline architecture:

- A metric-specific clean measurement, taken from a consistent physiological window.
- A recent trend, usually covering around 3 to 7 valid comparable readings.
- A working personal reference, normally around 28 days.
- An extended reference of up to around 60 days when the metric is sparse or naturally slow-moving.
- Metric-specific centre and spread calculations rather than one universal formula.
- Separate handling for circular time, bounded measurements and skewed measurements.
- Explicit context for illness, travel, altitude, menstrual cycle, device changes and non-wear.
- A confidence layer based on valid observations and source quality, shown alongside the result rather than affecting the result in any way.
- Slow adaptation after a persistent change, with short anomalies prevented from immediately rewriting the reference.

The exact windows and thresholds still need validation against our own users and score outcomes. They are a starting structure, not proven constants. A useful baseline remembers who the user normally is while still noticing genuine change: too wide and it says nothing, too responsive and it explains away the signal we wanted to find.

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